

- 11-5 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11-5.

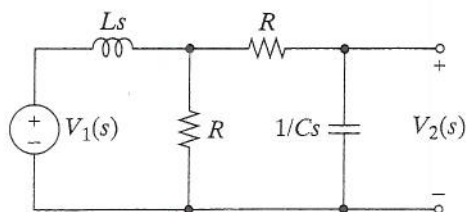


FIGURE P11-5

- 11-9 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11-9. Locate the poles and zeros of the transfer function.

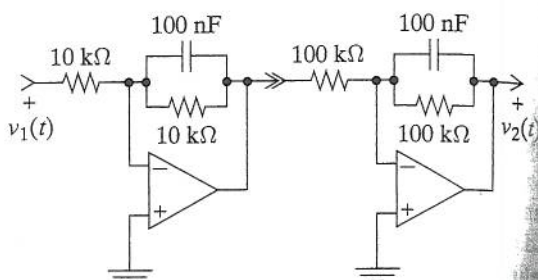


FIGURE P11-9

- 11-10 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11-10. Locate the poles and zeros of the transfer function.

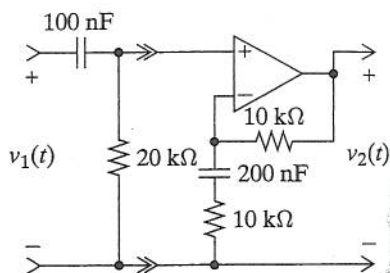


FIGURE P11-10

- 11-29 The circuit in Figure P11-29 is in the steady state with $i_1(t) = 5 \cos 1000t$ mA, $R = 1$ kΩ, $L = 2$ H, and $C = 500$ nF. Find $i_{2ss}(t)$. Repeat with $i_1(t) = 10 \cos 2000t$ mA.

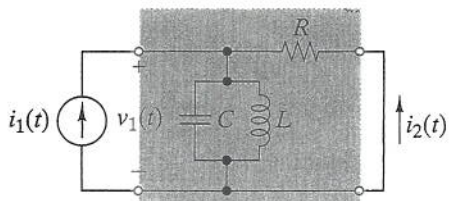


FIGURE P11-29

- 11-54 Design a circuit to realize the transfer function below using only resistors, capacitors, and inductors (no OP AMPS allowed). Scale the circuit so that all inductors are 50 mH or less.

$$T_V(s) = \frac{s^2}{(s + 2000)(s + 4000)}$$

- 11-59 It is claimed that both circuits in Figure P11-59 realize the transfer function

$$T_V(s) = \frac{\pm 1000 s}{(s + 1000)(s + 4000)}$$

- Verify that both circuits realize the specified $T_V(s)$.
- Which circuit would you choose if the output must drive a 1-kΩ load?
- Which circuit would you choose if the input comes from a 50-Ω source?
- It is further claimed that connecting the two circuits in cascade produces an overall transfer function of $[T_V(s)]^2$ no matter which circuit is the first stage and which is the second stage. Do you agree or disagree? Explain.

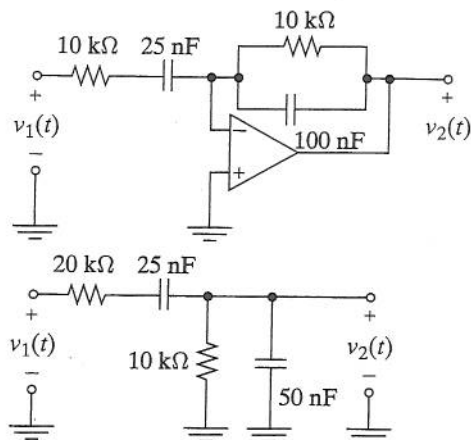


FIGURE P11-59

- 12-6 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-6. What type of gain response does the circuit have? What is the passband gain? Select values of R and C so that the cutoff frequency is 5 kHz.

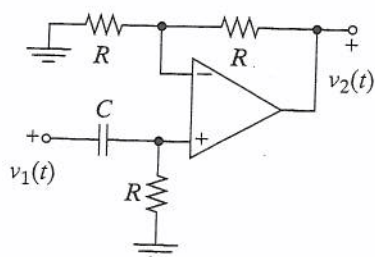


FIGURE P12-6

- 12-9 The transfer function of a first-order circuit is

$$T(s) = \frac{2}{10^{-2} + 20/s}$$

- Identify the type of gain response. Find the cutoff frequency and the passband gain.
- Sketch the straight-line gain response. Use the straight-line gain to estimate the gain at $\omega = 0.5\omega_c$, ω_c , and $2\omega_c$.

- 12-29 Show that the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-29 has a bandstop filter characteristic. Derive expressions relating the passband gains, the notch frequency, and the cutoff frequencies to R , L , and C .

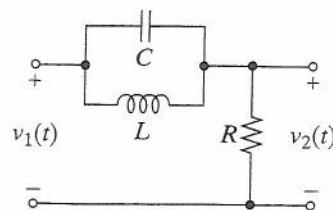


FIGURE P12-29

- 12-30 Figure P12-30 shows an RLC filter with an input current and an output voltage. The purpose of this problem is to determine the filter type using informal circuit analysis. Use the element impedances and basic analysis tools to find the magnitude of the output voltage $|V_2(j\omega)|$ at $\omega = 0$, $\omega = \infty$, and $\omega = \omega_0 = 1/\sqrt{LC}$. What is the filter type?

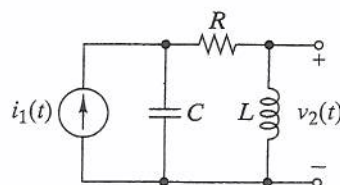


FIGURE P12-30